

Speed, Distance & Time



What Is Speed?

Speed tells us how fast something moves.

If you run quickly, your speed is high!

If you walk slowly, your speed is low.

Speed = how much distance you cover in a certain time



What Is Distance?

Distance is how far you travel from one place to another.
We measure it in metres (m) or kilometres (km).

Example: Your home to school might be 2 km!

What Is Time?

Time tells us how long it takes to go somewhere.
We measure time in seconds, minutes, or hours.

Example: It takes 10 minutes to walk to school!



The Magic Formula!

$$\text{Speed} = \text{Distance} / \text{Time}$$

If you go far in less time, your speed is faster!

If you go a short distance in more time, your speed is slower!

Remember: $S = D / T$

Example 1: Walking to School

Riya walks 2 km
She takes 30 minutes
What is her speed?

Speed = Distance / Time
Speed = 2 km / 30 min
Speed = 4 km per hour

Example 2: Cycling to the Park

Arjun cycles 6 km
He takes 30 minutes
What is his speed?

$\text{Speed} = \text{Distance} / \text{Time}$
 $\text{Speed} = 6 \text{ km} / 30 \text{ min}$
 $\text{Speed} = 12 \text{ km per hour}$



Example 3: Running Race

Priya runs 100 metres
She takes 20 seconds
What is her speed?

$\text{Speed} = \text{Distance} / \text{Time}$
 $\text{Speed} = 100 \text{ m} / 20 \text{ sec}$
 $\text{Speed} = 5 \text{ m per second}$

Quiz Time! Question 1

A car travels 60 km in 2 hours. What is its speed?

A) 20 km/hr

B) 30 km/hr

C) 40 km/hr

Answer: B) 30 km/hr

Quiz Time! Question 2

You run 200 metres in 40 seconds. What is your speed?

A) 3 m/sec

B) 5 m/sec

C) 8 m/sec

Answer: B) 5 m/sec

Quiz Time! Question 3

A bus takes 3 hours to travel 150 km. What is its speed?

A) 40 km/hr

B) 50 km/hr

C) 60 km/hr

Answer: B) 50 km/hr

Real Life: School Bus

Your school bus travels at 40 km per hour.
It covers 20 km from your stop to school.
How long does it take?

$$\text{Time} = \text{Distance} / \text{Speed} = 20/40 = 30 \text{ minutes!}$$



Real Life: Train Journey

A train travels from Delhi to Jaipur.
Distance = 280 km, Time = 4 hours.
What is the train's speed?

$$\text{Speed} = 280 / 4 = 70 \text{ km per hour!}$$



What Did We Learn?

Speed tells us how fast we go

Distance tells us how far we travel

Time tells us how long it takes

The magic formula: $\text{Speed} = \text{Distance} / \text{Time}$

We can solve real-life problems with this!

Great Job!

You learned Speed, Distance & Time today!
Do you have any questions?



Bonus: Speed Challenge!

Try this at home!

1. Mark 10 metres on the ground
2. Run from start to finish
3. Ask a friend to time you with a stopwatch
4. Calculate: $\text{Speed} = 10 \text{ m} / \text{your time}$

Share your speed with your classmates!